NLO Gluon Production: DGLAP, Running-Coupling, JIMWLK and BFKL Pieces

An exact rewriting of the NLO single-gluon spectrum and its separation into evolution terms

$d_\perp = 2-\epsilon$ (so $2/\epsilon = 1/\hat\epsilon$ for $D = 4-2\hat\epsilon$), $\overline{\rm MS}$ scale $\mu \equiv \mu_{\overline{MS}}$, pure glue ($n_f = 0$).

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0. Starting expression

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}-1\Big]\\
&\quad\times\Big[\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)/\xi}\Big(\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\Big)\Big]\,\mathcal{O}\\
&+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\Big[\log\frac{k^+}{\Lambda}+\log\Big(\frac{\vee}{k^+}-1\Big)\Big]\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[\cdots\Big]\,\mathcal{O}\\
&+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{k^+/\vee}^{k^+/(k^++\Lambda)}d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)/\xi}\Big[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-d_\perp\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\Big]\mathcal{O}
\end{aligned}
$$

where $[\cdots]$ in the second line is the same bracket as in the first, and $\mathcal{O} = [U^{e'c}(y')-U^{e'c}(x')][U^{ec}(y)-U^{ec}(x)]\,\rho^{e'}(x')\rho^{e}(x)$.

> In the original expression the last Wilson line reads $U^{ce}(x)$. It is taken here to be $U^{ec}(x)$; otherwise that factor is $U^\dagger(x)$.

1. Notation

$$
r\equiv y'-y,\qquad \mathcal{N}\equiv\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{1}{k^+},\qquad \frac{g^2N_c}{2\pi^2}=2\bar\alpha_s,\quad \bar\alpha_s=\frac{\alpha_sN_c}{\pi},\qquad a\equiv\frac{k^+}{\vee},\quad b\equiv\frac{k^+}{k^++\Lambda},
$$

$$
\mathcal{A}=\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2},\qquad \mathcal{B}=\frac{(x-y)\cdot r\,(x'-y')\cdot r}{(x-y)^2(x'-y')^2\,r^2},\qquad \Phi_W(q)\equiv\int_{x,y,x',y'}e^{-iq\cdot r}\,W\,\mathcal{O}.
$$

$\Phi_{\mathcal{A}}(q)$ is the LO structure (amplitude times conjugate) evaluated at transverse momentum q.

2. Two exact preliminaries

2.1 The bracket L

Pull out $d_\perp/2$:

$$
L\equiv-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{r^2\mu^2}{4}-1=-\frac{d_\perp}{2}\Big[\frac{2}{\epsilon}+\log\frac{r^2\mu^2}{c_0^2}\Big]-1,\qquad c_0\equiv2e^{-\gamma_E}.
$$

Using $d_\perp = 2-\epsilon$, the $O(\epsilon)$ part of $d_\perp/\epsilon$ gives $+1$, which cancels the -1 exactly:

$$
\boxed{L=-\frac{2}{\epsilon}-\log\frac{r^2\mu^2}{c_0^2}+O(\epsilon).}
$$

The $O(\epsilon)$ terms can be dropped because L only multiplies finite quantities: the ξ-integrals at fixed $\vee,\Lambda$, and a $g\to gg$ splitting function that is ϵ-independent. Equivalently,

$$
L=4\pi\int\frac{d^{d_\perp}q}{(2\pi)^{d_\perp}}\,\frac{e^{iq\cdot r}}{q^2}\;\simeq\;\frac{1}{\pi}\int d^{d_\perp}z\,\frac{(z-y)\cdot(z-y')}{(z-y)^2(z-y')^2}.
$$

So the pole is a final-state collinear one: relative transverse momentum $q\to0$, i.e. a large separation of the unobserved gluon at z, with no Wilson line $U(z)$.

2.2 The blue bracket

$$
\log\frac{k^+}{\Lambda}+\log\Big(\frac{\vee}{k^+}-1\Big)=\log\frac{\vee-k^+}{\Lambda}=\log\frac{k^+}{\Lambda}+\log\frac{\vee}{k^+}+\log\Big(1-\frac{k^+}{\vee}\Big).
$$

The last piece is $O(k^+/\vee)$; it is not a rapidity log. With a,b as defined in Section 1 below, $b/(1-b)=k^+/\Lambda$ and $a/(1-a)=k^+/(\vee-k^+)$ hold exactly, so the blue bracket is exactly

$$
\log\frac{k^+}{\Lambda}+\log\Big(\frac{\vee}{k^+}-1\Big)=\int_{k^+/\vee}^{k^+/(k^++\Lambda)}\frac{d\xi}{\xi(1-\xi)}=\int_a^b d\xi\Big[\frac{1}{1-\xi}+\frac{1}{\xi}\Big].
$$

The $1/(1-\xi)$ part gives the $\xi\to1$ (soft) end and the $1/\xi$ part gives the $\xi\to k^+/\vee$ end, both with the phase frozen at its $\xi=1$ value $e^{-ik\cdot r}$. This matters for the BFKL term (Section 7.2).

3. Exact rewriting

All three lines share the same prefactor $2\bar\alpha_s\mathcal{N}$. In the third line $d_\perp\to2$, because it has no pole.

$$\frac{d^3N_{\rm NLO}}{d^3k} = 2\bar\alpha_s\mathcal{N}\Big\{\underbrace{\int_a^1\frac{d\xi}{\xi^2}\,p(\xi)\,\Phi_{\mathcal{A}L}\big(\tfrac{k}{\xi}\big)}_{\text{(I)}} + \underbrace{\log\frac{\vee-k^+}{\Lambda}\,\Phi_{\mathcal{A}L}(k)}_{\text{(II)}} + \underbrace{\int_a^b d\xi\,\frac{1-\xi}{\xi}\,\Phi_{\mathcal{A}-2\mathcal{B}}\big(\tfrac{k}{\xi}\big)}_{\text{(III)}}\Big\}$$

$$p(\xi) = \xi(1-\xi) + \frac{1-\xi}{\xi} + \frac{\xi}{(1-\xi)_+}.$$

The full pure-glue splitting function, with $\beta_0 = \frac{11}{3}N_c$, is

$$P_{gg}(\xi) = 2N_c\,p(\xi) + \frac{\beta_0}{2}\,\delta(1-\xi) \qquad\Longrightarrow\qquad p(\xi) = \frac{P_{gg}(\xi)}{2N_c} - \frac{11}{12}\,\delta(1-\xi).$$

The momentum sum rule confirms the coefficient:

$$\int_0^1 d\xi\,\xi\,p(\xi) = -\tfrac{3}{2} + \tfrac{1}{2} + \tfrac{1}{12} = -\tfrac{11}{12}, \qquad \text{so} \qquad \int_0^1 d\xi\,\xi\,P_{gg} = 0.$$

Adding and subtracting the $\delta(1-\xi)$ term in (I) gives the separation below.

4. DGLAP: line (I), completed to the full P_{gg}

$$\frac{d^3N^{\rm DGLAP}}{d^3k} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{4\pi^2}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,P_{gg}(\xi)\int_{x,y,x',y'} e^{-ik\cdot r/\xi}\,\mathcal{A}\Big[-\frac{2}{\epsilon} - \log\frac{r^2\mu^2}{c_0^2}\Big]\mathcal{O}$$

- Here $(1-\xi)_+$ is the standard plus distribution on $[0,1]$.
- If the plus distribution in the original expression is defined on $[k^+/\vee, 1]$, the $\log(1-k^+/\vee)$ piece of the blue bracket converts it to the $[0,1]$ one exactly: $\int_a^1 d\xi\,\frac{f(\xi)-f(1)}{1-\xi} + f(1)\log(1-a)$, with $f(\xi) = e^{-ik\cdot r/\xi}/\xi$ and $f(1) = e^{-ik\cdot r}$.
- If it was already the $[0,1]$ one, that piece instead stays as an $O(k^+/\vee)$ leftover (Section 8).
- This is the timelike $g\to g$ fragmentation convolution: $\frac{d\xi}{\xi^3}$ times $\frac{dN_{\rm LO}}{d^3p} \propto \frac{1}{p^+}$ at $p^+ = k^+/\xi, p = k/\xi$.
- The $-2/\epsilon$ is removed by the $\overline{\rm MS}$ counterterm of $D_g(\xi,\mu^2)$. The natural scale is $\mu = c_0/r$, i.e. $\mu\sim k$ in momentum space.
- The $(1-\xi)/\xi$ term of P_{gg} is not log-free at the lower limit: its $\xi\to k^+/\vee$ end contains the BFKL log. That piece is extracted in Section 7. The DGLAP term proper is what remains, $\big[\mathrm{DGLAP}-\mathrm{BFKL}\big]$, which stays finite as $\vee\to\infty$.

5. Running coupling: the subtracted $\delta(1-\xi)$ piece

$$\frac{d^3N^{\rm RC}}{d^3k} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{4\pi^2}\frac{1}{k^+}\frac{11N_c}{6}\int_{x,y,x',y'} e^{-ik\cdot r}\,\mathcal{A}\Big[\frac{2}{\epsilon} + \log\frac{r^2\mu^2}{c_0^2}\Big]\mathcal{O} = \frac{\alpha_s\beta_0}{2\pi}\mathcal{N}\Big[\frac{2}{\epsilon}\,\Phi_{\mathcal{A}}(k) + \Phi_{\mathcal{A}\log(r^2\mu^2/c_0^2)}(k)\Big]$$

- In pure dimensional regularization the on-shell self-energy is scaleless, so its $1/\epsilon_{\rm UV}$ and $1/\epsilon_{\rm IR}$ cancel. That lets this pole be read as the UV pole removed by coupling renormalization of the LO.
- The log then makes the LO coupling run to the dipole scale:

$$\alpha_s(\mu^2) \;\to\; \alpha_s\big(c_0^2/r^2\big) = \alpha_s(\mu^2)\Big[1 + \frac{\alpha_s\beta_0}{4\pi}\log\frac{r^2\mu^2}{c_0^2}\Big].$$

- With quarks, $\beta_0 \to \frac{11N_c-2n_f}{3}$, and the $g\to q\bar q$ real channel must also be present.

6. JIMWLK: the term proportional to $\log(k^+/\Lambda)$

$$\frac{d^3N^{\rm JIMWLK}}{d^3k} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2N_c}{2\pi^2}\frac{1}{k^+}\,\log\frac{k^+}{\Lambda}\int_{x,y,x',y'} e^{-ik\cdot r}\,\mathcal{A}\Big[-\frac{2}{\epsilon} - \log\frac{r^2\mu^2}{c_0^2}\Big]\mathcal{O}$$

This comes from the soft end $\xi\to 1$: the unobserved gluon has $p^+ - k^+ = k^+(1-\xi)/\xi$, which reaches the cutoff Λ at $\xi = b$.

7. BFKL: the term proportional to $\log(\vee/k^+)$

The BFKL term comes from the $\xi\to k^+/\vee$ end of the $(1-\xi)/\xi$ term in line (I). There the measured gluon is soft relative to its parent ($p^+ = k^+/\xi$ approaches $\vee$). It does *not* come from the blue bracket; see Section 7.2.

7.1 Derivation

Step 1: isolate the term. The $(1-\xi)/\xi$ part of line (I) is

$$T \equiv 2\bar\alpha_s\mathcal{N}\int_a^1 d\xi\,\frac{1-\xi}{\xi^3}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)/\xi}\,\mathcal{A}(x,y,x',y')\,L(y'-y)\,\mathcal{O}(y,y').$$

Step 2: rescale the dipole. Put $y' = y + \xi s$, so $d^2y' = \xi^2 d^2 s$ and the phase becomes $e^{-ik\cdot s}$:

$$T = 2\bar{\alpha}_s \mathcal{N} \int_a^1 d\xi \, \frac{1-\xi}{\xi} \int_{x,y,x',s} e^{-ik\cdot s} \, \mathcal{A}(x,y,x',y+\xi s)\, L(\xi s)\, \mathcal{O}(y,y+\xi s).$$

Step 3: take $\xi \to 0$. Only this end can produce $\log(1/a)$. In that limit $\mathcal{A} \to \mathcal{A}(x,y,x',y)$ and $\mathcal{O} \to \mathcal{O}(y,y)$: the parent gluon crosses the target at the same point in the amplitude and in the conjugate. Split

$$L(\xi s) = -\frac{2}{\epsilon} - \log \frac{\xi^2 \mu^2}{c_0^2} - \log s^2.$$

The first two terms do not depend on s. With them the s-integral is the LO structure at momentum k/ξ, $\xi^{-2}\Phi_{\mathcal{A}}(k/\xi)$, which falls off much faster than ξ^2. So the $1/\epsilon$ pole and the $\log \mu^2$ give *no* $\log(\vee/k^+)$. Only $-\log s^2$ contributes.

Step 4: Fourier transform. For $k \neq 0$, using $\nabla^2 \log s^2 = 4\pi\delta^2(s)$ and $\int d^2w\, e^{ik\cdot(w-z)} \frac{(w-z)_i}{(w-z)^2} = 2\pi i \, \frac{k_i}{k^2}$,

$$\int d^2 s \, e^{-ik\cdot s}\big(-\log s^2\big) = \frac{4\pi}{k^2} = \frac{1}{\pi}\int_{w,w'} e^{-ik\cdot(w'-w)} \, \frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2} \qquad (\text{any } z).$$

This is the soft-gluon emission kernel of the measured gluon at w, w' off the hard gluon at z. Equivalently, it uses $L(r) \simeq \frac{1}{\pi}\int_z \frac{(z-y)\cdot(z-y')}{(z-y)^2(z-y')^2}$ from Section 2.1.

Step 5: do the ξ integral. $\int_a^1 d\xi \, \frac{1-\xi}{\xi} = \log \frac{\vee}{k^+} - 1 + \frac{k^+}{\vee}$. Rename $y \to z$ and keep the log.

$$\frac{d^3N^{\mathrm{BFKL}}}{d^3k} = \frac{2\bar{\alpha}_s\mathcal{N}}{\pi} \log\frac{\vee}{k^+} \int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)} \frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2} \frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2} \big[U^{e'c}(z) - U^{e'c}(x')\big]\big[U^{ec}(z) - U^{ec}(x)\big]\rho^{e'}(x')\rho^{e}(x)$$

$$\frac{2\bar{\alpha}_s\mathcal{N}}{\pi} = \frac{g^2 N_c}{2\pi^2} \cdot \frac{g^2}{4\pi^2} \frac{1}{(2\pi)^3} \frac{1}{k^+} \cdot \frac{1}{\pi} = \frac{1}{(2\pi)^3} \frac{g^4 N_c}{8\pi^5} \frac{1}{k^+}.$$

This is the expected BFKL term, prefactor included, with $\int_x = \int d^2x$. Its corrections are $O(1)$ terms with no $\log(\vee/k^+)$, and they belong in the finite remainder.

7.2 The explicit $\log(\vee/k^+)$ in line (II) is not the BFKL term

- Its Wilson lines sit at the measured-gluon points $y \neq y'$, not at one common point z. It also carries the collinear pole $-2/\epsilon$, which the BFKL term does not.
- Section 2.2 shows it is $\int_a \frac{d\xi}{\xi}\, \Phi_{\mathcal{A}L}(k)$, a $1/\xi$ end with the phase frozen at $\xi = 1$. That is not the correct $\xi \to 0$ limit of the $(1-\xi)/\xi$ term, which is the structure derived above.
- At large k, $\Phi_{\mathcal{A}L}(k) \to \frac{4\pi}{k^2}\int_{x,x',z} \mathcal{A}\,\mathcal{O}\big|_{y=y'=z}$. So the line (II) piece tends to exactly the BFKL term and would double it.
- To end up with only the expected BFKL term, the piece $2\bar{\alpha}_s\mathcal{N}\log\frac{\vee}{k^+}\Phi_{\mathcal{A}L}(k)$ must cancel against contributions not shown here, for example virtual graphs with the same upper cutoff $\vee$. Otherwise it is double counting from the step that produced $\log(\vee/k^+ - 1)$; check how that log was generated.

8. What is left over

8.1 The $\log(1-k^+/\vee)$ term

$$2\bar{\alpha}_s\mathcal{N}\, \log\Big(1 - \frac{k^+}{\vee}\Big)\, \Phi_{\mathcal{A}L}(k).$$

Depending on the plus-prescription convention, this either becomes the endpoint term in Section 4 or stays as a power correction.

8.2 Line (III) is finite and has no rapidity log

- **Upper end.** The factor $(1-\xi)$ vanishes at $\xi \to 1$, so setting $b \to 1$ costs only $\int_b^1 (1-\xi)\, d\xi = O(\Lambda^2/k^{+2})$. No $\log(k^+/\Lambda)$.
- **Lower end.** As $r \to 0$, the r-dependence of $\mathcal{A} - 2\mathcal{B}$ is the traceless combination $(\delta_{ij} - 2\hat{r}_i\hat{r}_j)$, whose angular average vanishes. Its Fourier transform is

  $$\int d^2r\, e^{-iq\cdot r}\, (\delta_{ij} - 2\hat{r}_i\hat{r}_j) = -\frac{4\pi}{q^2}\, (\delta_{ij} - 2\hat{q}_i\hat{q}_j),$$

  so $\Phi_{\mathcal{A}-2\mathcal{B}}(k/\xi) \sim \xi^2/k^2$. The integrand $\frac{1-\xi}{\xi}\,\Phi$ then goes like ξ, and setting $a \to 0$ costs only $O(k^{+2}/\vee^2)$. No $\log(\vee/k^+)$.

$$\frac{d^3N^{\mathrm{finite}}}{d^3k} = \frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2N_c}{2\pi^2}\frac{1}{k^+}\int_0^1 d\xi\, \frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot r/\xi}\big[\mathcal{A}-2\mathcal{B}\big]\mathcal{O} + O\Big(\frac{\Lambda^2}{k^{+2}}, \frac{k^{+2}}{\vee^2}\Big)$$

This is the genuine NLO impact-factor term.

8.3 Bookkeeping

Original piece	Equals
Line (I)	[DGLAP − BFKL] + BFKL + RC
Line (II)	JIMWLK + [$\log(\vee/k^+)\,\Phi_{\mathcal{A}L}(k)$ piece, Section 7.2] + $\log(1-k^+/\vee)$ term
Line (III)	finite remainder

The separation is exact up to $O(\epsilon)$.

9. Things to check

1. **A factor of 2 in normalization.** Relative to $\mathcal{N}\Phi_{\mathcal{A}}$, the DGLAP coefficient above is $(\alpha_s/\pi)P_{gg}$ and the RC coefficient is $\alpha_s\beta_0/2\pi$. Both are the textbook values, $(\alpha_s/2\pi)P_{gg}$ and $\alpha_s\beta_0/4\pi$, only if the LO is

$$\frac{dN_{\rm LO}}{d^3k} = 2\,\mathcal{N}\,\Phi_{\mathcal{A}}(k).$$

   The two checks agree with each other. If the LO is $\mathcal{N}\Phi_{\mathcal{A}}$, the NLO prefactor has a stray factor of 2, for example from the identical-gluon factor $\frac{1}{2}$.
2. **The JIMWLK and BFKL pieces carry L, including its $1/\epsilon$.** They are soft–collinear double-log terms: a rapidity log times the final-state collinear pole. In the full NLO, $\frac{1}{\epsilon}\log\frac{k^+}{\Lambda}$ should cancel against virtual corrections with the same cutoff Λ; the $U(z)$ terms of the evolution kernel are not in the expression shown. Do not also absorb these poles into DGLAP, or they will be double counted. The finite evolution kernels are the $\log(r^2\mu^2/c_0^2)$ parts.
3. **Where the $\log(\vee/k^+)$ lives.** The BFKL term comes from the lower limit of line (I), so that limit cannot be set to 0. The explicit $\log(\vee/k^+ - 1)$ in line (II) has to cancel against something else, or it double counts (Section 7.2).